

Route G-R5: Acquired Clock Data and an External-Reference Observation Model

An empirical eligibility boundary and two measurable contrasts

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Abstract

Public E2/E3 comparison data were acquired and audited, including 2,324,201 records from the 2025 ROCIT release. They do not include the paired bath-state, independent radiometry and correction ledger needed for the requested two-temperature test. That empirical task remains open. Separately, we specify a second Yb⁺ E3 clock outside the changed environment and derive both differential and common target-reference observables. A thermal-response-cancelling composite is useful, but reference drift and common instrumental shifts remain exactly degenerate with an unexplained common signal unless independently constrained. No universal time law is fitted and no new experimental effect is claimed.

1 What the acquired data can and cannot establish

Two primary releases were obtained with checksums and source attribution:

- PTB’s 2021 release [1] contains 11 published E3/E2 ratio points, their times and uncertainty components. Its ratio column is an *absolute ratio offset*, not a fractional offset. The associated campaign used two clock systems. No paired thermal calibration is supplied.
- ROCIT’s September 2025 release [2] contains 34 selected same-package E2/E3 comparator files from the 2022 campaign: 2,324,201 rows, all marked valid, no nonfinite values or duplicate timestamps. Its columns are time, comparator output, validity and two systematic uncertainties. It supplies neither two-temperature assignments nor independent ion-site radiometry and a per-run BBR correction ledger.

Original release bytes are not the same as uncorrected atom/servo data. The complete small PTB archive and one exact ROCIT day plus metadata are preserved; every selected ROCIT member is hashed in the full audit. No thermal slope is extracted from these unsuitable records. This is a limitation of the audited releases, not a claim that qualifying data exist nowhere. The later public ROCIT deposit also supersedes the paper’s earlier generic “on request” access statement [3].

The ROCIT format [4] defines $\Delta_{A \rightarrow B} = (\hat{\nu}_B - \rho_{BA}^0 \hat{\nu}_A) / s_B$. For this pair A is E3 and B E2, opposite to our preferred E3/E2 ratio. Exact conversion requires its reference convention, not merely negating a column. Valid records need not be independent samples.

2 Specify the outside reference, including its response

Let S be one target Yb^+ ion, interleaved on E2 and E3 inside a separately controlled radiation enclosure. Let R be a second, independent Yb^+ E3 clock in a separate monitored enclosure outside the changed bath. Compare both target outputs to R using a frequency comb and phase-stabilized optical paths. This is a specified arrangement, not hardware constructed here. Existing separate-clock and external-clock comparisons support the architecture [3, 5], not our proposed thermal experiment's performance.

Retain G-R4's ordinary electric-dipole response at *both* sites:

$$b_i^X = -\frac{1}{2h\epsilon_0\nu_i^0} \mathcal{P} \int_0^\infty \Delta\alpha_i(\omega) u_\omega^X d\omega, \quad B_i^X = \log(1 + b_i^X), \quad X = S, R. \quad (1)$$

For ideal thermal radiation $u^X = a_{\text{rad}} T_X^4$; at leading static order $b_i^X = \beta_i u^X$. A real multi-surface bath generally needs its spectrum and geometry, not one unexamined wall temperature. Motional sideband thermometry does not measure radiation temperature.

Allow a candidate common local factor $F_X > 0$ without choosing a law for it. Write the measured logarithmic ratio as

$$q_i = \log \left(\frac{\nu_i^S / \nu_3^R}{\nu_i^0 / \nu_3^0} \right), \quad (2)$$

$$q_i = \chi_S - \chi_R + B_i^S - B_3^R + g_{SR} + \ell_i + A_i^S - A_3^R + n_i, \quad \chi_X = \log F_X. \quad (3)$$

A denotes the remaining ordinary *log-frequency* corrections, ℓ_i the path/comb contribution and n_i measurement noise. Using logs makes multiplicative factors additive; at first order A is the usual fractional nuisance shift. No reference term is set to zero simply because the reference is outside the enclosure. In a weak-field, slow-motion convention one ordinary contribution is

$$g_{SR} = \frac{\Phi_S - \Phi_R}{c^2} - \frac{v_S^2 - v_R^2}{2c^2}, \quad (4)$$

with additional transfer effects assigned to ℓ_i to avoid double counting. Fixed heights remove a static offset, not an unknown changing one. Conventional spacetime is assumed; no identification with Chen's E_d or completion of the finite G-R1–G-R3 constraints is implied.

For hot-minus-cold contrast Δ , independently correcting the ordinary terms yields

$$z_i = \Delta q_i - \Delta B_i^S + \Delta B_3^R - \Delta g_{SR} - \Delta \ell_i - \Delta A_i^S + \Delta A_3^R = \Delta(\chi_S - \chi_R) + \Delta n_i. \quad (5)$$

Real measurements average with sensitivity functions $W_i(t)$. Reference noise cancels in $q_3 - q_2$ only for matched effective windows: otherwise it leaves $-\int (W_3 - W_2) q_R(t) dt$. Interleaving alone does not guarantee cancellation; dead times and servo response matter.

3 Two contrasts and an exact identifiability obstruction

Suppress already evaluated ordinary terms for clarity. To first order, the two external comparisons have the structure

$$\begin{pmatrix} y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 & \beta_2 \\ 1 & \beta_3 \end{pmatrix} \begin{pmatrix} \chi \\ \Delta u_S \end{pmatrix} - \beta_3 \Delta u_R \mathbf{1} + c \mathbf{1} + \epsilon, \quad \chi = \Delta(\chi_S - \chi_R). \quad (6)$$

Here c represents an unevaluated common path, gravity or reference drift term. The differential observable is

$$D = y_3 - y_2 = (\beta_3 - \beta_2) \Delta u_S + \epsilon_3 - \epsilon_2. \quad (7)$$

Both χ and a shared reference term cancel. The composite

$$C = w_2 y_2 + w_3 y_3, \quad (8)$$

$$w_2 = \frac{\beta_3}{\beta_3 - \beta_2}, \quad w_3 = -\frac{\beta_2}{\beta_3 - \beta_2}, \quad w_2 + w_3 = 1, \quad w_2 \beta_2 + w_3 \beta_3 = 0 \quad (9)$$

instead gives $C = \chi - \beta_3 \Delta u_R + c + w^T \epsilon$. With G-R4's independently pinned inputs,

$$(w_2, w_3) \simeq (-0.16004, 1.16004). \quad (10)$$

This cancels the leading target T^4 response, not all dynamic or other shifts. Negative weighting can increase noise. The design matrix has determinant $\beta_3 - \beta_2 \neq 0$: with reference effects controlled, two external channels can distinguish a common change from the selected thermal response. But solving two channels for two unknowns leaves *no residual test*. Independent radiometry is needed to test the thermal law, rather than infer the bath from that law and call it verified.

Obstruction. If c is unconstrained, the transformation $(\chi, c) \mapsto (\chi + \lambda, c - \lambda)$ leaves all observations unchanged. Thus even this external comparison measures an unexplained common residual, not uniquely a time effect. A factor common to both environments cancels as before. A third ratio constructed from the same two channels supplies only a closure identity, not another independent equation; an independently measured link can diagnose instrumental closure.

For independently calibrated corrections in Eq. (5), the generalized least-squares common estimate and its variance are

$$\hat{\chi} = \frac{\mathbf{1}^T V^{-1} z}{\mathbf{1}^T V^{-1} \mathbf{1}}, \quad \text{Var } \hat{\chi} = (\mathbf{1}^T V^{-1} \mathbf{1})^{-1}. \quad (11)$$

This is an estimator specification, not a fit performed here. Shared reference error contributes $\sigma_R^2 \mathbf{1} \mathbf{1}^T$ to V ; it cancels from the differential variance but remains a floor in the common estimate. Correction covariance and sensitivity to the same calibration inputs must also be propagated, rather than counted as independent noise.

4 Numerical control requirements and the empirical handoff

For the illustrative target change $300 \rightarrow 310$ K, G-R4’s static coefficients predict

$$(\Delta b_2^S, \Delta b_3^S) \simeq (-7.337 \cdot 10^{-17}, -1.012 \cdot 10^{-17}), \quad D \simeq 6.325 \cdot 10^{-17}. \quad (12)$$

These are calculations, not observations. With fixed nominal composite weights, the uncertainty from independently calibrated polarizabilities is

$$\delta C = \Delta u_S(w_2 \delta \beta_2 + w_3 \delta \beta_3), \quad \sigma_C \in [|a_2 - a_3|, a_2 + a_3], \quad a_i = |w_i| \sigma_{\beta_i} |\Delta u_S|. \quad (13)$$

Unknown correlation gives σ_C between $2.17 \cdot 10^{-18}$ and $2.59 \cdot 10^{-18}$ from these inputs alone. This is not a confidence interval or an achieved total uncertainty. The retained historical calibrations do not support a sub- 10^{-18} common-mode claim. For scale, a reference bath change $300 \rightarrow 300.1$ K contributes about $-9.64 \cdot 10^{-20}$ to its E3 frequency, while a 1 cm height change gives $g\Delta h/c^2 \simeq 1.09 \cdot 10^{-18}$. These are conditional sensitivities, not asserted apparatus fluctuations.

Use settled low–high–high–low blocks with matched effective centers $t = (-3, -1, 1, 3)\tau$ and contrast weights $(-1, 1, 1, -1)/2$. Then $\sum w = \sum wt = 0$ and the hot-minus-cold response has unit gain. However $\sum wt^2 = -8\tau^2$: nonlinear drift survives, as does heater-correlated bias. Reversed ordering, monitored reference cross-talk and a dummy-heater/electrical-pickup control are useful discriminators, not proofs of cancellation. No particular dwell time or hardware precision is asserted without the thermal/servo response.

The next empirical input must supply uncorrected comparator/servo records, bath-state assignments, independent radiation calibration at both sites, every applied correction and its covariance, synchronized validity/averaging windows, and reference/path/gravity controls. The machine-readable contract is `EXTERNAL_REFERENCE_CARD.json`. No researcher has been contacted or new experiment performed. Obtaining those records remains open; the acquisition audit and reference-model derivation are bounded-done. The 46 new checks verify algebra and source eligibility, not evidence for a common time effect.

References

- [1] R. Lange et al., PTB comparison data (2021), doi:10.7795/720.20211028; associated paper: Phys. Rev. Lett. 126, 011102.
- [2] T. Lindvall et al., ROCIT campaign dataset (2025), doi:10.5281/zenodo.17107693.
- [3] T. Lindvall et al., *Coordinated international comparisons between optical clocks connected via fiber and satellite links*, Optica 12, 843 (2025).
- [4] INRIM, *Optical link data format*, source specification, accessed 24 September 2026.
- [5] M. Steinel et al., *Evaluation of a $^{88}\text{Sr}^+$ Optical Clock with a Direct Measurement of the Blackbody Radiation Shift and Determination of the Clock Frequency*, Phys. Rev. Lett. 131, 083002 (2023).